## a Cross-channel freshness

**GUI mutation**

Emitted / partial input
Changed desktop
Unknown mutation

**Invalidate**

Working snapshots
Hidden-element exposure cache
Find continuation cursors

**Retain**

App / window / element reference registries
Delivered-tree ledger
Eligible diff snapshots

**AX full / screenshot image** — becomes current GUI frame → **GUI layout lease**

**AX mutation** → **GUI layout lease**

Recheck before coordinate input

## b Action guards

**AX target reference**

Bound element + window

**AX target guard**

Target reference

```
button "OK" [ref=r75]
```

Live element




Resolve identity, per action

pass → **AX action + feedback**

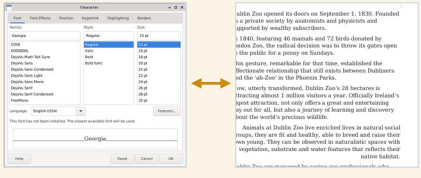
reject

**GUI screenshot**

Private layout lease

**GUI layout guard**

Screenshot layout ↔ Current layout



Check layout at target points

pass → **GUI input + screenshot**

reject `gui_frame_stale`

## c Recovery follows delivery evidence

**Uncertain preparation?**

yes → **Stop**

No replay, observation or later batch action

no

| Schema error | Known non-delivery | Unknown effect | Step or wait failure |
|---|---|---|---|
| `invalid_argument` | `stale_ref`<br>`gui_frame_stale` | `effect_unknown`<br>`window_changed` | `wait_timeout` |
| Correct the call | AX: observe/find<br>GUI: screenshot | Inspect before retry | Observe, then revise |